



```
In [15]: # Numpy Crash Course
```

```
In [16]: import numpy as np
```

```
In [17]: np.__version__
```

```
Out[17]: '2.3.5'
```

```
In [18]: import sys  
sys.version
```

```
Out[18]: '3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1  
929 64 bit (AMD64)]'
```

```
In [19]: ## Creating Arrays
```

```
In [20]: my_list = [0,1,2,3,4,5]  
my_list
```

```
Out[20]: [0, 1, 2, 3, 4, 5]
```

```
In [21]: type(my_list)
```

```
Out[21]: list
```

```
In [22]: arr = np.array(my_list)
```

```
In [23]: arr
```

```
Out[23]: array([0, 1, 2, 3, 4, 5])
```

```
In [24]: type(arr)
```

```
Out[24]: numpy.ndarray
```

```
In [25]: type(my_list)
```

```
Out[25]: list
```

```
In [28]: # numpy arrange
```

```
In [27]: np. # we learn important function
```

```
Cell In[27], line 1  
    np. # we learn important function  
      ^  
SyntaxError: invalid syntax
```

```
In [29]: np.arange(15)
```

```
Out[29]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14])
```

```
In [30]: np.arange(3.0)
```

```
Out[30]: array([0., 1., 2.])
```

```
In [31]: np.arange(10)
```

```
Out[31]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [32]: np.arange(0, 5)
```

```
Out[32]: array([0, 1, 2, 3, 4])
```

```
In [33]: np.arange(10,20)
```

```
Out[33]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [34]: np.arange(20,10) # 1st arg < 2nd arg
```

```
Out[34]: array([], dtype=int64)
```

```
In [35]: np.arange(-20,10)
```

```
Out[35]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
                -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                 6,  7,  8,  9])
```

```
In [36]: np.arange(-16,10)
```

```
Out[36]: array([-16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4,
                -3, -2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,  9])
```

```
In [37]: np.arange(-20,-10)
```

```
Out[37]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11])
```

```
In [38]: np.arange(30,20) # 1st arg always be < then 2nd arg
```

```
Out[38]: array([], dtype=int64)
```

```
In [39]: np.arange()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[39], line 1
----> 1 np.arange()

TypeError: arange() requires stop to be specified.
```

```
In [40]: np.arange(10,30,5) # 10- starting from 30- end point 5 - step count
```

```
Out[40]: array([10, 15, 20, 25])
```

```
In [41]: np.arange(0,10,3)
```

```
Out[41]: array([0, 3, 6, 9])
```

```
In [42]: np.arange(10,30,5,8)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[42], line 1  
----> 1 np.arange(10,30,5,8)  
  
TypeError: Cannot interpret '8' as a data type
```

```
In [43]: np.zeros(3) # parameter tuning
```

```
Out[43]: array([0., 0., 0.])
```

```
In [44]: np.zeros(5, dtype=int) # hyperparameter tuning
```

```
Out[44]: array([0, 0, 0, 0, 0])
```

```
In [45]: np.zeros((2,2), dtype=int)
```

```
Out[45]: array([[0, 0],  
               [0, 0]])
```

```
In [46]: zero = np.zeros([2,2])  
print(zero)  
print(type(zero))
```

```
[[0. 0.]  
 [0. 0.]  
<class 'numpy.ndarray'>
```

```
In [47]: zero = np.zeros([2,2])  
print(zero)  
  
print('####')  
  
print(type(zero))
```

```
[[0. 0.]  
 [0. 0.]  
####  
<class 'numpy.ndarray'>
```

```
In [48]: np.zeros((2,10))
```

```
Out[48]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [49]: np.zeros((2,2))
```

```
Out[49]: array([[0., 0.],
               [0., 0.]])
```

```
In [50]: np.zeros((3,3))
```

```
Out[50]: array([[0., 0., 0.],
               [0., 0., 0.],
               [0., 0., 0.]])
```

```
In [51]: np.zeros((10,30))
```

```
Out[51]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [52]: np.zeros((5,10)) # by default large -- will give row & 2nd arg - columns
```

```
Out[52]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [53]: n = (6,7)
         n1 = (6,8)
         print(np.zeros(n1)) # parameter tuning
```

```
[[0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
```

```
In [54]: print(np.zeros(n,dtype=int)) ## hyperparameter tuning
```

```
[[0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0]]
```

In [55]: `n`

Out[55]: (6, 7)

In [56]: `n1`

Out[56]: (6, 8)

In [57]: `print(np.zeros(n1))`

```
[[0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]]
```

In [58]: `# numpy with ones`

In [59]: `np.ones(3)`

Out[59]: `array([1., 1., 1.])`

In [60]: `np.ones(4, dtype=int)`

Out[60]: `array([1, 1, 1, 1])`

In [61]: `np.ones(4)`

Out[61]: `array([1., 1., 1., 1.])`

In [62]: `n`

Out[62]: (6, 7)

In [63]: `np.ones(n)`

Out[63]: `array([[1., 1., 1., 1., 1., 1., 1.],
 [1., 1., 1., 1., 1., 1., 1.],
 [1., 1., 1., 1., 1., 1., 1.],
 [1., 1., 1., 1., 1., 1., 1.],
 [1., 1., 1., 1., 1., 1., 1.],
 [1., 1., 1., 1., 1., 1., 1.]])`

In [64]: `np.ones((5,4),dtype=int) # by default 5- rows & 4 - columns`

```
Out[64]: array([[1, 1, 1, 1],
               [1, 1, 1, 1],
               [1, 1, 1, 1],
               [1, 1, 1, 1],
               [1, 1, 1, 1]])
```

```
In [65]: np.
```

```
Cell In[65], line 1
  np.
    ^
SyntaxError: invalid syntax
```

```
In [66]: np.twos((2,3))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[66], line 1
----> 1 np.twos((2,3))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

```
In [ ]: np.three(2,3)
```

```
In [67]: np.ones(2)
```

```
Out[67]: array([1., 1.])
```

```
In [68]: np.ones((2,4))
```

```
Out[68]: array([[1., 1., 1., 1.],
               [1., 1., 1., 1.]])
```

```
In [69]: np.ones((6,10),dtype = int)
```

```
Out[69]: array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])
```

```
In [70]: np.twos((2,4))
```

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[70], line 1
----> 1 np.twos((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'

```

In [71]: `np.`

```

Cell In[71], line 1
    np.
    ^
SyntaxError: invalid syntax

```

In [72]: `np.three((2,4))`

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[72], line 1
----> 1 np.three((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'

```

In [73]: `range(5)`

Out[73]: `range(0, 5)`

In [74]: `r = range(5)`
`r`

Out[74]: `range(0, 5)`

In [75]: `for i in r:`
 `print(i)`

```

0
1
2
3
4

```

In [76]: `list(range(5))`

Out[76]: [0, 1, 2, 3, 4]

```
In [77]: range(1,10)
```

Out[77]: range(1, 10)

```
In [78]: list(range(1,10))
```

Out[78]: [1, 2, 3, 4, 5, 6, 7, 8, 9]

```
In [79]: list(range(1,10,3))
```

Out[79]: [1, 4, 7]

```
In [80]: y = list(range(12))
y
```

Out[80]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

```
In [81]: # Random value
```

```
In [82]: rand(3,2)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[82], line 1
----> 1 rand(3,2)

NameError: name 'rand' is not defined
```

```
In [83]: rand(3,2)
random.rand(3,2)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[83], line 1
----> 1 rand(3,2)
      2 random.rand(3,2)

NameError: name 'rand' is not defined
```

```
In [84]: np.random.rand(5)
```

Out[84]: array([0.24793152, 0.94036474, 0.90056223, 0.55458024, 0.85585868])

```
In [85]: np.rand(4)
```



```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[85], line 1  
----> 1 np.rand(4)  
  
File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)  
    792     import numpy.char as char  
    793     return char.chararray  
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")  
  
AttributeError: module 'numpy' has no attribute 'rand'
```

```
In [86]: np.random.rand(2,4)
```

```
Out[86]: array([[0.05579826, 0.6418894 , 0.0719777 , 0.58957724],  
               [0.06874176, 0.68546352, 0.50206731, 0.79987714]])
```

```
In [87]: np.random.randint(2,4)
```

```
Out[87]: 2
```

```
In [88]: np.random.randint(2,20) # 2nd argument is exclusive
```

```
Out[88]: 13
```

```
In [89]: np.random.randint(0,1)
```

```
Out[89]: 0
```

```
In [90]: np.random.randint(10,20,5)
```

```
Out[90]: array([15, 17, 13, 17, 12], dtype=int32)
```

```
In [91]: np.random.randint(1,6,4)
```

```
Out[91]: array([2, 5, 1, 3], dtype=int32)
```

```
In [92]: np.random.rand(3)
```

```
Out[92]: array([0.82232946, 0.38536802, 0.57916916])
```

```
In [93]: np.random.randint(1)
```

```
Out[93]: 0
```

```
In [94]: np.random.randint(30,20,10)
```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[94], line 1
----> 1 np.random.randint(30,20,10)

File numpy/random/mtrand.pyx:794, in numpy.random.mtrand.RandomState.randint()

File numpy/random/_bounded_integers.pyx:1425, in numpy.random._bounded_integers._rand_int32()

ValueError: low >= high

```

```
In [95]: np.random.randint(-30,20,10)
```

```
Out[95]: array([-10, -3, -14, 12,  0, -8,  6, -17, -20, -4], dtype=int32)
```

```
In [96]: np.random.randint(20,30,10)
```

```
Out[96]: array([20, 20, 29, 29, 25, 22, 25, 27, 28, 29], dtype=int32)
```

```
In [97]: np.random.randint(5,9)  #GET THE VALUE <=1 & >=5
```

```
Out[97]: 7
```

```
In [98]: np.random.randint(10,21,3)
```

```
Out[98]: array([15, 14, 12], dtype=int32)
```

```
In [99]: np.random.randint(1,12,10)
```

```
Out[99]: array([11,  9, 11,  9,  9,  9,  1, 11,  3,  4], dtype=int32)
```

```
In [100... np.random.randint(10,40,(10,10)) #generre the element 10 -30 with 4*4 mtri
```

```
Out[100... array([[39, 29, 16, 23, 17, 20, 15, 26, 16, 18],
                  [18, 33, 24, 31, 39, 16, 24, 12, 25, 32],
                  [18, 18, 25, 29, 17, 22, 19, 32, 25, 17],
                  [21, 14, 21, 38, 36, 27, 24, 31, 15, 33],
                  [28, 15, 18, 33, 35, 12, 21, 31, 17, 15],
                  [39, 15, 30, 27, 30, 19, 24, 14, 27, 14],
                  [22, 35, 18, 33, 33, 12, 24, 36, 29, 37],
                  [35, 35, 39, 28, 18, 29, 27, 19, 27, 33],
                  [24, 25, 28, 22, 24, 31, 14, 10, 25, 37],
                  [28, 13, 37, 16, 32, 22, 31, 34, 33, 23]], dtype=int32)
```

```
In [101... np.random.randint(1,100,(12,12)) #generre the element 10 -30 with 4*4 mtri
```

```
Out[101...] array([[42, 14, 50,  4, 20,  5,  9,  8,  3, 35, 53, 91],
                  [23, 31, 26, 63, 21, 52, 65, 40,  3, 39, 36, 19],
                  [74, 19, 23, 58, 80, 35, 27, 76, 75, 57, 58, 15],
                  [12, 52,  9, 34, 33, 18, 90, 54, 56, 82, 79, 56],
                  [40, 78, 88, 19, 24, 58, 88, 47, 29, 46, 38, 12],
                  [67,  8, 87, 82, 35, 38, 41, 97, 17,  7, 33, 65],
                  [57, 83, 83, 88, 52, 30, 58, 95, 34, 60, 33, 87],
                  [52,  5, 96, 56, 77, 15, 84, 70,  1, 50, 90, 76],
                  [61, 81, 87, 29,  7, 11, 19, 89, 75, 41, 63,  4],
                  [53, 75, 95, 83, 71, 36, 53, 57, 29,  3, 41, 37],
                  [37, 39, 32, 32, 82, 60, 59, 32, 88, 68, 86,  6],
                  [48, 20, 42, 70, 81, 77, 90, 36, 34, 47,  7, 82]], dtype=int32)
```

```
In [102...] np.arange(1,13).reshape(3,4)
```

```
Out[102...] array([[ 1,  2,  3,  4],
                  [ 5,  6,  7,  8],
                  [ 9, 10, 11, 12]])
```

```
In [103...] np.arange(1,13).reshape(12, 1)
```

```
Out[103...] array([[ 1],
                  [ 2],
                  [ 3],
                  [ 4],
                  [ 5],
                  [ 6],
                  [ 7],
                  [ 8],
                  [ 9],
                  [10],
                  [11],
                  [12]])
```

```
In [104...] b = np.random.randint(10,20,(5,4))
b
```

```
Out[104...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]], dtype=int32)
```

```
In [105...] type(b)
```

```
Out[105...] numpy.ndarray
```

```
In [106...] b
```

```
Out[106... array([[16, 18, 12, 17],
                [18, 15, 19, 16],
                [10, 10, 15, 12],
                [12, 12, 15, 18],
                [10, 18, 11, 16]]], dtype=int32)
```

```
In [107... b[:]
```

```
Out[107... array([[16, 18, 12, 17],
                [18, 15, 19, 16],
                [10, 10, 15, 12],
                [12, 12, 15, 18],
                [10, 18, 11, 16]]], dtype=int32)
```

```
In [108... b[1:3]
```

```
Out[108... array([[18, 15, 19, 16],
                [10, 10, 15, 12]]], dtype=int32)
```

```
In [109... b
```

```
Out[109... array([[16, 18, 12, 17],
                [18, 15, 19, 16],
                [10, 10, 15, 12],
                [12, 12, 15, 18],
                [10, 18, 11, 16]]], dtype=int32)
```

```
In [110... b[1,2]
```

```
Out[110... np.int32(19)
```

```
In [111... b
```

```
Out[111... array([[16, 18, 12, 17],
                [18, 15, 19, 16],
                [10, 10, 15, 12],
                [12, 12, 15, 18],
                [10, 18, 11, 16]]], dtype=int32)
```

```
In [112... b[1,3]
```

```
Out[112... np.int32(16)
```

```
In [113... b[1,-1]
```

```
Out[113... np.int32(16)
```

```
In [114... b
```

```
Out[114...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]]), dtype=int32)
```

```
In [115...] b[2:3]
```

```
Out[115...] array([[10, 10, 15, 12]], dtype=int32)
```

```
In [116...] b
```

```
Out[116...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]]), dtype=int32)
```

```
In [117...] b[0:-2]
```

```
Out[117...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12]], dtype=int32)
```

```
In [118...] b
```

```
Out[118...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]]), dtype=int32)
```

```
In [119...] b[-5, -3]
```

```
Out[119...] np.int32(18)
```

```
In [120...] b
```

```
Out[120...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]]), dtype=int32)
```

```
In [121...] b[-4, 2]
```

```
Out[121...] np.int32(19)
```

```
In [122...] np.random.randint(10, 20, (4, 4))
```

```
Out[122...] array([[19, 13, 17, 10],
                  [11, 13, 18, 11],
                  [11, 15, 19, 14],
                  [16, 10, 10, 17]], dtype=int32)
```

```
In [123...] b
```

```
Out[123...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]], dtype=int32)
```

```
In [124...] b[-4,-2]
```

```
Out[124...] np.int32(19)
```

```
In [125...] b
```

```
Out[125...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]], dtype=int32)
```

```
In [126...] b[-4:2]
```

```
Out[126...] array([[18, 15, 19, 16]], dtype=int32)
```

```
In [127...] b[:]
```

```
Out[127...] array([[16, 18, 12, 17],
                  [18, 15, 19, 16],
                  [10, 10, 15, 12],
                  [12, 12, 15, 18],
                  [10, 18, 11, 16]], dtype=int32)
```

```
In [128...] ## Operations
```

```
In [129...] a = np.random.randint(10,20,10)
a
```

```
Out[129...] array([17, 19, 16, 13, 13, 15, 10, 19, 14, 15], dtype=int32)
```

```
In [130...] id(a)
```

```
Out[130...] 2514096302352
```

```
In [131...] arr
```

```
Out[131...] array([0, 1, 2, 3, 4, 5])
```

```
In [132... arr2 = np.random.randint(0,100,(10,10))
```

```
In [133... arr2
```

```
Out[133... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
          [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
          [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
          [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
          [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
          [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
          [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
          [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
          [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
          [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]], dtype=int32)
```

```
In [134... arr
```

```
Out[134... array([0, 1, 2, 3, 4, 5])
```

```
In [135... arr[:]
```

```
Out[135... array([0, 1, 2, 3, 4, 5])
```

```
In [136... arr
```

```
Out[136... array([0, 1, 2, 3, 4, 5])
```

```
In [137... arr[:4]
```

```
Out[137... array([0, 1, 2, 3])
```

```
In [138... arr2[:]
```

```
Out[138... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
          [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
          [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
          [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
          [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
          [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
          [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
          [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
          [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
          [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]], dtype=int32)
```

```
In [139... arr2[0:5]
```

```
Out[139... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
          [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
          [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
          [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
          [98, 73, 18, 15, 17, 65, 92, 86, 45, 41]], dtype=int32)
```

```
In [140... arr2
```

```
Out[140... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
          [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
          [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
          [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
          [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
          [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
          [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
          [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
          [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
          [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]]], dtype=int32)
```

```
In [141... arr2[1,4]
```

```
Out[141... np.int32(79)
```

```
In [142... arr2
```

```
Out[142... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
          [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
          [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
          [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
          [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
          [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
          [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
          [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
          [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
          [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]]], dtype=int32)
```

```
In [143... arr2[-5,5]
```

```
Out[143... np.int32(8)
```

```
In [144... arr2[-5,-5]
```

```
Out[144... np.int32(8)
```

```
In [145... arr2
```

```
Out[145... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
          [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
          [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
          [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
          [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
          [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
          [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
          [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
          [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
          [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]]], dtype=int32)
```

```
In [146... arr2[-5,-5]
```


Out[146... np.int32(8)

In [147... arr2

```
Out[147... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
                [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
                [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
                [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
                [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
                [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]]), dtype=int32)
```

In [148... arr2[-1,-2]

Out[148... np.int32(22)

In [149... arr2

```
Out[149... array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
                [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
                [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
                [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
                [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
                [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]]), dtype=int32)
```

In [150... arr2[::-1]

```
Out[150... array([[11, 38, 89, 50, 23, 71, 51, 56, 22, 27],
                [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
                [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
                [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
                [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
                [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                [44, 68,  9, 12, 34, 93,  8, 86, 43, 11]]), dtype=int32)
```

In [151... arr2[::-1]

```
Out[151...] array([[11, 38, 89, 50, 23, 71, 51, 56, 22, 27],
                  [99, 60, 91, 40, 6, 5, 2, 61, 11, 32],
                  [ 0, 22, 76, 57, 68, 86, 1, 49, 42, 7],
                  [81, 35, 10, 1, 79, 65, 55, 79, 74, 37],
                  [54, 98, 0, 58, 39, 8, 5, 71, 74, 72],
                  [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                  [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                  [44, 68, 9, 12, 34, 93, 8, 86, 43, 11]], dtype=int32)
```

```
In [152...] arr2
```

```
Out[152...] array([[44, 68, 9, 12, 34, 93, 8, 86, 43, 11],
                  [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                  [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                  [54, 98, 0, 58, 39, 8, 5, 71, 74, 72],
                  [81, 35, 10, 1, 79, 65, 55, 79, 74, 37],
                  [ 0, 22, 76, 57, 68, 86, 1, 49, 42, 7],
                  [99, 60, 91, 40, 6, 5, 2, 61, 11, 32],
                  [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]], dtype=int32)
```

```
In [153...] arr2[:, :-2]
```

```
Out[153...] array([[11, 38, 89, 50, 23, 71, 51, 56, 22, 27],
                  [ 0, 22, 76, 57, 68, 86, 1, 49, 42, 7],
                  [54, 98, 0, 58, 39, 8, 5, 71, 74, 72],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [90, 13, 14, 50, 79, 97, 71, 73, 11, 68]], dtype=int32)
```

```
In [154...] arr2
```

```
Out[154...] array([[44, 68, 9, 12, 34, 93, 8, 86, 43, 11],
                  [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                  [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                  [54, 98, 0, 58, 39, 8, 5, 71, 74, 72],
                  [81, 35, 10, 1, 79, 65, 55, 79, 74, 37],
                  [ 0, 22, 76, 57, 68, 86, 1, 49, 42, 7],
                  [99, 60, 91, 40, 6, 5, 2, 61, 11, 32],
                  [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]], dtype=int32)
```

```
In [155...] arr2[:, :-3]
```

```
Out[155...] array([[11, 38, 89, 50, 23, 71, 51, 56, 22, 27],
                  [81, 35, 10, 1, 79, 65, 55, 79, 74, 37],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [44, 68, 9, 12, 34, 93, 8, 86, 43, 11]], dtype=int32)
```

```
In [156...] arr2
```

```
Out[156...] array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
                  [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                  [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                  [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
                  [81, 35, 10,  1, 79, 65, 55, 79, 74, 37],
                  [ 0, 22, 76, 57, 68, 86,  1, 49, 42,  7],
                  [99, 60, 91, 40,  6,  5,  2, 61, 11, 32],
                  [11, 38, 89, 50, 23, 71, 51, 56, 22, 27]], dtype=int32)
```

```
In [157...] arr2[:-3]
```

```
Out[157...] array([[44, 68,  9, 12, 34, 93,  8, 86, 43, 11],
                  [90, 13, 14, 50, 79, 97, 71, 73, 11, 68],
                  [86, 67, 14, 20, 20, 44, 52, 80, 39, 59],
                  [46, 32, 76, 78, 95, 58, 91, 98, 53, 99],
                  [98, 73, 18, 15, 17, 65, 92, 86, 45, 41],
                  [54, 98,  0, 58, 39,  8,  5, 71, 74, 72],
                  [81, 35, 10,  1, 79, 65, 55, 79, 74, 37]], dtype=int32)
```

```
In [158...] arr
```

```
Out[158...] array([0, 1, 2, 3, 4, 5])
```

```
In [159...] arr.max()
```

```
Out[159...] np.int64(5)
```

```
In [160...] arr.min()
```

```
Out[160...] np.int64(0)
```

```
In [161...] arr
```

```
Out[161...] array([0, 1, 2, 3, 4, 5])
```

```
In [162...] arr.mean()
```

```
Out[162...] np.float64(2.5)
```

```
In [163...] arr
```

```
Out[163...] array([0, 1, 2, 3, 4, 5])
```

```
In [164...] arr.median()
```

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[164], line 1  
----> 1 arr.median()  
  
AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

```
In [165... from numpy import *  
a = array([1,2,3,4,9])  
median(a)
```

```
Out[165... np.float64(3.0)
```

```
In [166... Without work on import* can you please find the median, mode)
```

```
Cell In[166], line 1  
Without work on import* can you please find the median, mode)  
^  
SyntaxError: unmatched ')'
```

```
In [ ]: arr
```

```
In [167... arr.reshape(3,2)
```

```
Out[167... array([[0, 1],  
                [2, 3],  
                [4, 5]])
```

```
In [168... arr.reshape(6,1)
```

```
Out[168... array([[0],  
                [1],  
                [2],  
                [3],  
                [4],  
                [5]])
```

```
In [169... arr.reshape(1,6)
```

```
Out[169... array([[0, 1, 2, 3, 4, 5]])
```

```
In [170... arr
```

```
Out[170... array([0, 1, 2, 3, 4, 5])
```

```
In [171... arr.reshape(2,4)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[171], line 1  
----> 1 arr.reshape(2,4)  
  
ValueError: cannot reshape array of size 6 into shape (2,4)
```

```
In [172... arr
```

```
Out[172... array([0, 1, 2, 3, 4, 5])
```

```
In [173... arr.reshape(2,3,order='C')
```

```
Out[173... array([[0, 1, 2],  
                [3, 4, 5]])
```

```
In [174... arr.reshape(2,3,order='F') # print element with fortran
```

```
Out[174... array([[0, 2, 4],  
                [1, 3, 5]])
```

```
In [175... arr.reshape(2,3,order='A') # A almost give you c type output
```

```
Out[175... array([[0, 1, 2],  
                [3, 4, 5]])
```

```
In [176... arr
```

```
Out[176... array([0, 1, 2, 3, 4, 5])
```

```
In [177... arr.reshape(2,3)
```

```
Out[177... array([[0, 1, 2],  
                [3, 4, 5]])
```

```
In [178... arr.reshape(1,4)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[178], line 1  
----> 1 arr.reshape(1,4)  
  
ValueError: cannot reshape array of size 6 into shape (1,4)
```

```
In [179... arr.reshape(1,6)
```

```
Out[179... array([0, 1, 2, 3, 4, 5])
```

```
In [180... arr.reshape(6,1)
```

```
Out[180... array([[0],  
                [1],  
                [2],  
                [3],  
                [4],  
                [5]])
```

```
In [181... arr.reshape(2,6)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[181], line 1  
----> 1 arr.reshape(2,6)  
  
ValueError: cannot reshape array of size 6 into shape (2,6)
```

```
In [182... arr.reshape(3,3)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[182], line 1  
----> 1 arr.reshape(3,3)  
  
ValueError: cannot reshape array of size 6 into shape (3,3)
```

```
In [183... arr
```

```
Out[183... array([0, 1, 2, 3, 4, 5])
```

```
In [184... arr.reshape(3,2)
```

```
Out[184... array([[0, 1],  
                [2, 3],  
                [4, 5]])
```

```
In [185... ## Indexing
```

```
In [186... mat = np.arange(0,100).reshape(10,10)  
mat
```

```
Out[186... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [187... row = 4  
col = 5
```

```
In [188... col
```

```
Out[188... 5
```

```
In [189... row
```

```
Out[189... 4
```

```
In [190... mat
```

```
Out[190... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [191... mat[6] # default it represent to rows
```

```
Out[191... array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
In [192... mat
```

```
Out[192... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [193... # With Slices
mat[:,col]
```

```
Out[193... array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
In [194... mat
```

```
Out[194... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [195... mat[row,:]
```

```
Out[195... array([40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [196... mat
```

```
Out[196...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [197...] mat
```

```
Out[197...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [198...] mat[:,]
```

```
Out[198...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [199...] mat[:,8]
```

```
Out[199...] array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [200...] mat
```

```
Out[200...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```



```
In [201... mat[:, -1]
```

```
Out[201... array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
In [202... mat
```

```
Out[202... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [203... row
```

```
Out[203... 4
```

```
In [204... col
```

```
Out[204... 5
```

```
In [205... mat[:, col]
```

```
Out[205... array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
In [206... mat
```

```
Out[206... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [207... mat[1, 4]
```

```
Out[207... np.int64(14)
```

```
In [208... mat
```

```
Out[208... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [209... mat[1:4]
```

```
Out[209... array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [210... mat[3:-3]
```

```
Out[210... array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [211... mat[0]
```

```
Out[211... array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [212... mat[6]
```

```
Out[212... array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
In [213... mat
```

```
Out[213... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [214... mat[6:]
```

```
Out[214... array([[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [215... mat[:6]
```

```
Out[215...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [216...] mat[5:7]
```

```
Out[216...] array([[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [217...] mat[0:10]
```

```
Out[217...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [218...] mat[0:10:3]
```

```
Out[218...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [219...] mat[0:10:3]
```

```
Out[219...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [220...] mat[4:]
```

```
Out[220...] array([[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [221...] mat[:, :-1]
```

```
Out[221... array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [222... mat[:, -3]
```

```
Out[222... array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [223... mat[:, -3]
```

```
Out[223... array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [224... mat[:, -5]
```

```
Out[224... array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])
```

```
In [225... mat[2:6]
```

```
Out[225... array([[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [226... mat[2:6, 2:4] # 1:5 --> only row part /// 1:3 -- it indicates only column parts
```

```
Out[226... array([[22, 23],
                [32, 33],
                [42, 43],
                [52, 53]])
```

```
In [227... mat[0, 1]
```

```
Out[227... np.int64(1)
```

```
In [228... mat[1, 6]
```

```
Out[228... np.int64(16)
```

```
In [229... mat[1:]
```

```
Out[229...] array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [230...] mat
```

```
Out[230...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [231...] mat[:6]
```

```
Out[231...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [232...] mat[0:1]
```

```
Out[232...] array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]])
```

```
In [233...] mat[1:2,2:4]
```

```
Out[233...] array([[12, 13]])
```

```
In [234...] mat[2:3,2:3]
```

```
Out[234...] array([[22]])
```

```
In [235...] mat[3:5,2:4]
```

```
Out[235...] array([[32, 33],
                  [42, 43]])
```

```
In [236...] mat[2:3,4:5]
```

```
Out[236...] array([[24]])
```

```
In [237... ## Masking
```

```
In [238... mat # we also called as filter
```

```
Out[238... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [239... id(mat)
```

```
Out[239... 2514098772112
```

```
In [240... mat
```

```
Out[240... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [241... mat[mat<50]
```

```
Out[241... array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
                34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [242... mat > 50
```

```
Out[242...] array([[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True]])
```

```
In [243...] mat[mat==50]
```

```
Out[243...] array([50])
```

```
In [244...] mat
```

```
Out[244...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [245...] mat == 50
```

```
Out[245...] array([[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[ True, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False]])
```

```
In [246...] a1 = mat[mat<50]
a1
```

```
Out[246...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [247...] a2 = mat[mat>50]
a2
```

```
Out[247...] array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

```
In [248...] a3 = mat[mat<=50]
a3
```

```
Out[248...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])
```

```
In [249...] a4 = mat[mat==50]
a4
```

```
Out[249...] array([50])
```

```
In [250...] # python program to generat otp
```

```
In [251...] import random

def generate_otp(length=4):
    """Generate a numeric OTP of a specified length."""
```



```

digits = '012345'
otp = ''.join(random.choice(digits) for _ in range(length))
return otp

# Example usage
otp_length = 4 # You can change this to any length you prefer
otp = generate_otp(otp_length)
print(f"Your OTP is: {otp}")

```

Your OTP is: 0050

```

In [252... def wish():
            print('good even')
            wish()

            def wish():
                print('good even')
                wish()

            def wish():
                print('good even')
                wish()

```

good even
good even
good even

```

In [253... list1=['a','b','g',1,5]
            print(list1.pop)

```

<built-in method pop of list object at 0x000002495BEF2B40>

```

In [254... x = [1, 2, 3]
            y = x.copy()
            x.append(4)
            print(x)

```

[1, 2, 3, 4]

```

In [255... ## Creating Arrays

```

```

In [256... my_list = [0,1,2,3,4,5]
            my_list

```

Out[256... [0, 1, 2, 3, 4, 5]

```

In [257... type(my_list)

```

Out[257... list

```

In [258... arr = np.array(my_list)

```

```

In [259... arr

```

```
Out[259... array([0, 1, 2, 3, 4, 5])
```

```
In [260... type(arr)
```

```
Out[260... numpy.ndarray
```

```
In [261... type(my_list)
```

```
Out[261... list
```

```
In [262... np. # we learn important function
```

```
Cell In[262], line 1
      np. # we learn important function
      ^
SyntaxError: invalid syntax
```

```
In [263... np.arange(15)
```

```
Out[263... array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14])
```

```
In [264... np.arange(3.0)
```

```
Out[264... array([0., 1., 2.])
```

```
In [265... np.arange(10)
```

```
Out[265... array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [266... np.arange(0, 5)
```

```
Out[266... array([0, 1, 2, 3, 4])
```

```
In [267... np.arange(10,20)
```

```
Out[267... array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [268... np.arange(20,10) # 1st arg < 2nd arg
```

```
Out[268... array([], dtype=int64)
```

```
In [269... np.arange(-20,10)
```

```
Out[269... array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
        -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
         6,  7,  8,  9])
```

```
In [270... np.arange(-16,10)
```

```
Out[270... array([-16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4,
        -3, -2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,  9])
```

```
In [271... np.arange(-20,-10)
```

```
Out[271... array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11])
```

```
In [272... np.arange(30,20) # 1st arg always be < then 2nd arg
```

```
Out[272... array([], dtype=int64)
```

```
In [273... ar = np.arange(-30,20)  
ar
```

```
Out[273... array([-30, -29, -28, -27, -26, -25, -24, -23, -22, -21, -20, -19, -18,  
        -17, -16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5,  
        -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,  
         9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [274... np.arange(10,10)
```

```
Out[274... array([], dtype=int64)
```

```
In [275... np.arange()
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[275], line 1  
----> 1 np.arange()  
  
TypeError: arange() requires stop to be specified.
```

```
In [276... np.arange(10,30,5) # 10- starting from 30- end point 5 - step count
```

```
Out[276... array([10, 15, 20, 25])
```

```
In [277... np.arange(0,10,3)
```

```
Out[277... array([0, 3, 6, 9])
```

```
In [278... np.arange(10,30,5,8)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[278], line 1  
----> 1 np.arange(10,30,5,8)  
  
TypeError: Cannot interpret '8' as a data type
```

```
In [279... np.zeros(3) # parameter tuning
```

```
Out[279... array([0., 0., 0.])
```

```
In [280... np.zeros(5, dtype=int) # hyperparameter tuning
```

```
Out[280... array([0, 0, 0, 0, 0])
```

```
In [281... np.zeros((2,2), dtype=int)
```

```
Out[281... array([[0, 0],  
                [0, 0]])
```

```
In [282... zero = np.zeros([2,2])  
print(zero)  
print(type(zero))
```

```
[[0. 0.]  
 [0. 0.]  
<class 'numpy.ndarray'>
```

```
In [283... zero = np.zeros([2,2])  
print(zero)  
  
print('####')  
  
print(type(zero))
```

```
[[0. 0.]  
 [0. 0.]  
####  
<class 'numpy.ndarray'>
```

```
In [284... np.zeros((2,10))
```

```
Out[284... array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
                [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [285... np.zeros((2,2))
```

```
Out[285... array([[0., 0.],  
                [0., 0.]])
```

```
In [286... np.zeros((3,3))
```

```
Out[286... array([[0., 0., 0.],  
                [0., 0., 0.],  
                [0., 0., 0.]])
```

```
In [287... np.zeros((10,30))
```


Out[292... (6, 8)

In [293... `print(np.zeros(n1))`

```
[[0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]]
```

In [294... `np.ones(3)`

Out[294... `array([1., 1., 1.])`

In [295... `np.ones(4, dtype=int)`

Out[295... `array([1, 1, 1, 1])`

In [296... `np.ones(4)`

Out[296... `array([1., 1., 1., 1.])`

In [297... `n`

Out[297... (6, 7)

In [298... `np.ones(n)`

```
array([[1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1.]])
```

In [299... `np.ones((5,4),dtype=int) # by default 5- rows & 4 - columns`

```
array([[1, 1, 1, 1],
       [1, 1, 1, 1],
       [1, 1, 1, 1],
       [1, 1, 1, 1],
       [1, 1, 1, 1]])
```

In [300... `np.`

```
Cell In[300], line 1
np.
^
SyntaxError: invalid syntax
```

In [301... `np.twos((2,3))`

In [307... `np.`

```
Cell In[307], line 1
  np.
    ^
SyntaxError: invalid syntax
```

In [308... `np.three((2,4))`

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[308], line 1
----> 1 np.three((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'
```

In [309... `range(5)`

Out[309... `range(0, 5)`

In [310... `r = range(5)`
`r`

Out[310... `range(0, 5)`

In [311... `for i in r:`
 `print(i)`

```
0
1
2
3
4
```

In [312... `list(range(5))`

Out[312... `[0, 1, 2, 3, 4]`

In [313... `list(range(1,10))`

Out[313... `[1, 2, 3, 4, 5, 6, 7, 8, 9]`

In [314... `list(range(1,10,3))`

Out[314... `[1, 4, 7]`

In [315... `y = list(range(12))`
`y`

Out[315... [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

In [316... `rand(3,2)`

```
-----
NameError                                Traceback (most recent call last)
Cell In[316], line 1
----> 1 rand(3,2)

NameError: name 'rand' is not defined
```

In [317... `rand(3,2)`
`random.rand(3,2)`

```
-----
NameError                                Traceback (most recent call last)
Cell In[317], line 1
----> 1 rand(3,2)
      2 random.rand(3,2)

NameError: name 'rand' is not defined
```

In [318... `np.random.rand(5)`

Out[318... `array([0.48169857, 0.16782151, 0.56010213, 0.18280454, 0.93724948])`

In [319... `np.rand(4)`

```
-----
AttributeError                            Traceback (most recent call last)
Cell In[319], line 1
----> 1 np.rand(4)

File ~\anaconda3\Lib\site-packages\numpy\_init_.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'rand'
```

In [320... `np.random.rand(2,4)`

Out[320... `array([[0.17417047, 0.85467222, 0.55995771, 0.75908793],`
 `[0.88359343, 0.03719931, 0.50175251, 0.77721586]])`

In [321... `np.random.rand(2,4)`

Out[321... `array([[0.72625097, 0.63307768, 0.91169167, 0.08821285],`
 `[0.94433089, 0.8280985 , 0.23663143, 0.07105873]])`

In [322... `np.random.randint(2,20) # 2nd argument is exclusive`

Out[322... 6

```
In [323... np.random.randint(0,1)
```

```
Out[323... 0
```

```
In [324... np.random.randint(0,1)
```

```
Out[324... 0
```

```
In [325... np.random.randint(10,20,5)
```

```
Out[325... array([12, 12, 13, 14, 17], dtype=int32)
```

```
In [326... np.random.randint(1,6,4)
```

```
Out[326... array([3, 4, 1, 5], dtype=int32)
```

```
In [327... np.random.rand(3)
```

```
Out[327... array([0.44626216, 0.95210783, 0.6982174 ])
```

```
In [328... np.random.randint(1)
```

```
Out[328... 0
```

```
In [329... np.random.randint(30,20,10)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[329], line 1  
----> 1 np.random.randint(30,20,10)  
  
File numpy/random/mtrand.pyx:794, in numpy.random.mtrand.RandomState.randint()  
  
File numpy/random/_bounded_integers.pyx:1425, in numpy.random._bounded_integers._rand_int32()  
  
ValueError: low >= high
```

```
In [330... np.random.randint(-30,20,10)
```

```
Out[330... array([ -2, -12,  3, -10,  8,  4, 12, -15,  0,  1], dtype=int32)
```

```
In [331... np.random.randint(20,30,10)
```

```
Out[331... array([22, 25, 24, 24, 29, 20, 25, 28, 29, 26], dtype=int32)
```

```
In [332... np.random.randint(5,9) #GET THE VALUE <=1 & >=5
```

```
Out[332... 5
```

```
In [333... np.random.randint(10,21,3)
```

```
Out[333... array([16, 12, 16], dtype=int32)
```

```
In [334... np.random.randint(1,12,10)
```

```
Out[334... array([10, 1, 1, 7, 4, 6, 6, 11, 4, 7], dtype=int32)
```

```
In [335... np.random.randint(10,40,(10,10)) #generre the element 10 -30 with 4*4 mtri
```

```
Out[335... array([[37, 16, 23, 17, 38, 15, 34, 15, 14, 34],
        [13, 36, 38, 26, 13, 30, 21, 29, 14, 32],
        [36, 26, 27, 12, 14, 31, 23, 10, 33, 39],
        [19, 12, 19, 28, 28, 33, 17, 27, 34, 38],
        [37, 30, 11, 13, 12, 36, 33, 33, 10, 27],
        [10, 10, 32, 17, 26, 39, 19, 15, 26, 12],
        [35, 16, 18, 21, 10, 13, 37, 33, 31, 25],
        [13, 35, 20, 13, 26, 12, 10, 19, 16, 31],
        [24, 36, 28, 25, 17, 39, 29, 31, 35, 37],
        [20, 12, 37, 35, 12, 13, 30, 14, 19, 20]], dtype=int32)
```

```
In [336... np.random.randint(1,100,(12,12)) #generre the element 10 -30 with 4*4 mtri
```

```
Out[336... array([[92, 21, 52, 8, 26, 92, 82, 97, 21, 72, 59, 5],
        [69, 43, 7, 46, 57, 50, 81, 7, 63, 15, 34, 81],
        [73, 40, 99, 26, 69, 62, 89, 95, 50, 4, 71, 72],
        [1, 99, 91, 33, 39, 6, 57, 50, 72, 34, 84, 42],
        [26, 89, 9, 95, 69, 49, 99, 91, 92, 84, 80, 84],
        [67, 75, 98, 94, 9, 31, 4, 85, 33, 48, 89, 65],
        [77, 30, 32, 56, 60, 56, 35, 2, 78, 83, 34, 94],
        [42, 92, 34, 89, 1, 3, 98, 24, 35, 16, 61, 4],
        [67, 14, 47, 45, 55, 70, 66, 74, 2, 94, 56, 34],
        [82, 87, 58, 28, 14, 42, 96, 75, 97, 48, 84, 70],
        [42, 51, 75, 53, 27, 12, 75, 59, 36, 85, 43, 85],
        [49, 44, 88, 27, 23, 14, 9, 80, 91, 65, 28, 99]], dtype=int32)
```

```
In [337... np.arange(1,13).reshape(3,4)
```

```
Out[337... array([[1, 2, 3, 4],
        [5, 6, 7, 8],
        [9, 10, 11, 12]])
```

```
In [338... np.arange(1,13).reshape(12, 1)
```

```
Out[338...] array([[ 1],
                  [ 2],
                  [ 3],
                  [ 4],
                  [ 5],
                  [ 6],
                  [ 7],
                  [ 8],
                  [ 9],
                  [10],
                  [11],
                  [12]])
```

```
In [339...] b = np.random.randint(10,20,(5,4))
b
```

```
Out[339...] array([[14, 15, 16, 18],
                  [14, 11, 10, 19],
                  [12, 18, 11, 13],
                  [17, 16, 11, 15],
                  [15, 10, 10, 18]], dtype=int32)
```

```
In [340...] type(b)
```

```
Out[340...] numpy.ndarray
```

```
In [341...] b
```

```
Out[341...] array([[14, 15, 16, 18],
                  [14, 11, 10, 19],
                  [12, 18, 11, 13],
                  [17, 16, 11, 15],
                  [15, 10, 10, 18]], dtype=int32)
```

```
In [342...] b[:]
```

```
Out[342...] array([[14, 15, 16, 18],
                  [14, 11, 10, 19],
                  [12, 18, 11, 13],
                  [17, 16, 11, 15],
                  [15, 10, 10, 18]], dtype=int32)
```

```
In [343...] b[1:3]
```

```
Out[343...] array([[14, 11, 10, 19],
                  [12, 18, 11, 13]], dtype=int32)
```

```
In [344...] b
```

```
Out[344...] array([[14, 15, 16, 18],
                  [14, 11, 10, 19],
                  [12, 18, 11, 13],
                  [17, 16, 11, 15],
                  [15, 10, 10, 18]], dtype=int32)
```

```
In [345... b[1,2]
```

```
Out[345... np.int32(10)
```

```
In [346... b
```

```
Out[346... array([[14, 15, 16, 18],  
          [14, 11, 10, 19],  
          [12, 18, 11, 13],  
          [17, 16, 11, 15],  
          [15, 10, 10, 18]], dtype=int32)
```

```
In [347... b[1,3]
```

```
Out[347... np.int32(19)
```

```
In [348... b[1,-1]
```

```
Out[348... np.int32(19)
```

```
In [349... b
```

```
Out[349... array([[14, 15, 16, 18],  
          [14, 11, 10, 19],  
          [12, 18, 11, 13],  
          [17, 16, 11, 15],  
          [15, 10, 10, 18]], dtype=int32)
```

```
In [350... b[2:3]
```

```
Out[350... array([[12, 18, 11, 13]], dtype=int32)
```

```
In [351... b
```

```
Out[351... array([[14, 15, 16, 18],  
          [14, 11, 10, 19],  
          [12, 18, 11, 13],  
          [17, 16, 11, 15],  
          [15, 10, 10, 18]], dtype=int32)
```

```
In [352... b[0:-2]
```

```
Out[352... array([[14, 15, 16, 18],  
          [14, 11, 10, 19],  
          [12, 18, 11, 13]], dtype=int32)
```

```
In [353... b
```

```
Out[353... array([[14, 15, 16, 18],  
          [14, 11, 10, 19],  
          [12, 18, 11, 13],  
          [17, 16, 11, 15],  
          [15, 10, 10, 18]], dtype=int32)
```

```
In [354... b[0,2]
```

```
Out[354... np.int32(16)
```

```
In [355... b
```

```
Out[355... array([[14, 15, 16, 18],
          [14, 11, 10, 19],
          [12, 18, 11, 13],
          [17, 16, 11, 15],
          [15, 10, 10, 18]], dtype=int32)
```

```
In [356... b[-5,-3]
```

```
Out[356... np.int32(15)
```

```
In [357... b
```

```
Out[357... array([[14, 15, 16, 18],
          [14, 11, 10, 19],
          [12, 18, 11, 13],
          [17, 16, 11, 15],
          [15, 10, 10, 18]], dtype=int32)
```

```
In [358... b[-4,2]
```

```
Out[358... np.int32(10)
```

```
In [359... np.random.randint(10,20,(4,4))
```

```
Out[359... array([[16, 11, 19, 19],
          [19, 13, 16, 10],
          [15, 16, 11, 11],
          [14, 19, 17, 19]], dtype=int32)
```

```
In [360... b
```

```
Out[360... array([[14, 15, 16, 18],
          [14, 11, 10, 19],
          [12, 18, 11, 13],
          [17, 16, 11, 15],
          [15, 10, 10, 18]], dtype=int32)
```

```
In [361... b[-4,-2]
```

```
Out[361... np.int32(10)
```

```
In [362... b[-4:2]
```

```
Out[362... array([[14, 11, 10, 19]], dtype=int32)
```

```
In [363... b
```

```
Out[363... array([[14, 15, 16, 18],
                [14, 11, 10, 19],
                [12, 18, 11, 13],
                [17, 16, 11, 15],
                [15, 10, 10, 18]]], dtype=int32)
```

```
In [364... b[:]
```

```
Out[364... array([[14, 15, 16, 18],
                [14, 11, 10, 19],
                [12, 18, 11, 13],
                [17, 16, 11, 15],
                [15, 10, 10, 18]]], dtype=int32)
```

```
In [365... ## Operations
```

```
In [366... a = np.random.randint(10,20,10)
a
```

```
Out[366... array([15, 14, 13, 15, 16, 15, 16, 19, 17, 12], dtype=int32)
```

```
In [367... id(a)
```

```
Out[367... 2514100231056
```

```
In [368... arr
```

```
Out[368... array([0, 1, 2, 3, 4, 5])
```

```
In [369... arr2 = np.random.randint(0,100,(10,10))
```

```
In [370... arr2
```

```
Out[370... array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]], dtype=int32)
```

```
In [371... arr
```

```
Out[371... array([0, 1, 2, 3, 4, 5])
```

```
In [372... arr[:]
```

```
Out[372... array([0, 1, 2, 3, 4, 5])
```

```
In [373... arr
```

```
Out[373... array([0, 1, 2, 3, 4, 5])
```

```
In [374... arr[:4]
```

```
Out[374... array([0, 1, 2, 3])
```

```
In [375... arr2[:]
```

```
Out[375... array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [376... arr2[0:5]
```

```
Out[376... array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14]]), dtype=int32)
```

```
In [377... arr2
```

```
Out[377... array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [378... arr2[1,4]
```

```
Out[378... np.int32(1)
```

```
In [379... arr2
```



```
Out[379...] array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [380...] arr2[-5,5]
```

```
Out[380...] np.int32(79)
```

```
In [381...] arr2[-5,-5]
```

```
Out[381...] np.int32(79)
```

```
In [382...] arr2
```

```
Out[382...] array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [383...] arr2[-5,-5]
```

```
Out[383...] np.int32(79)
```

```
In [384...] arr2
```

```
Out[384...] array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [385...] arr2[-1,-2]
```

```
Out[385...] np.int32(32)
```

```
In [386... arr2
```

```
Out[386... array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
          [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
          [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
          [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
          [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
          [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
          [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
          [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
          [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
          [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [387... arr2[::-1]
```

```
Out[387... array([[72, 22, 99, 26, 97, 41, 76, 70, 32, 47],
          [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
          [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
          [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
          [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
          [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
          [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
          [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
          [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
          [11, 46, 29, 19,  3, 32, 86, 27,  8, 95]]), dtype=int32)
```

```
In [388... arr2
```

```
Out[388... array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
          [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
          [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
          [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
          [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
          [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
          [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
          [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
          [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
          [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]]), dtype=int32)
```

```
In [389... arr2[::-2]
```

```
Out[389... array([[72, 22, 99, 26, 97, 41, 76, 70, 32, 47],
          [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
          [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
          [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
          [21, 10, 38, 74,  1, 78, 61, 90,  4, 12]]), dtype=int32)
```

```
In [390... arr2[::2]
```

```
Out[390... array([[72, 22, 99, 26, 97, 41, 76, 70, 32, 47],
          [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
          [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
          [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
          [21, 10, 38, 74,  1, 78, 61, 90,  4, 12]]), dtype=int32)
```

```
In [391...] arr2
```

```
Out[391...] array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]], dtype=int32)
```

```
In [392...] arr2[::-3]
```

```
Out[392...] array([[72, 22, 99, 26, 97, 41, 76, 70, 32, 47],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [11, 46, 29, 19,  3, 32, 86, 27,  8, 95]], dtype=int32)
```

```
In [393...] arr2
```

```
Out[393...] array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49],
                  [14,  9, 47, 49, 45,  1, 30, 32, 33, 36],
                  [35, 92, 72, 49, 82, 62, 65, 79, 77, 62],
                  [72, 22, 99, 26, 97, 41, 76, 70, 32, 47]], dtype=int32)
```

```
In [394...] arr2[::-3]
```

```
Out[394...] array([[11, 46, 29, 19,  3, 32, 86, 27,  8, 95],
                  [21, 10, 38, 74,  1, 78, 61, 90,  4, 12],
                  [ 1, 61, 86,  7, 20, 51, 17, 12, 22, 63],
                  [ 0, 11,  5, 37, 85, 52, 13, 87, 48, 42],
                  [61, 87, 83, 39, 87, 67, 72,  9, 17, 14],
                  [65, 55, 91, 10, 97, 79, 60, 47, 73, 16],
                  [98, 55, 81,  0,  9, 26, 64,  5, 24, 49]], dtype=int32)
```

```
In [395...] arr
```

```
Out[395...] array([0, 1, 2, 3, 4, 5])
```

```
In [396...] arr.min()
```

```
Out[396...] np.int64(0)
```

```
In [397...] arr
```

Out[397... array([0, 1, 2, 3, 4, 5])

```
In [398... arr.mean()
```

Out[398... np.float64(2.5)

```
In [399... arr
```

Out[399... array([0, 1, 2, 3, 4, 5])

```
In [400... arr.median()
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[400], line 1
----> 1 arr.median()

AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

```
In [401... from numpy import *
a = array([1,2,3,4,9])
median(a)
```

Out[401... np.float64(3.0)

```
In [402... # Without work on import* can you please find the median, mode)
```

```
In [403... arr
```

Out[403... array([0, 1, 2, 3, 4, 5])

```
In [404... arr.reshape(3,2)
```

Out[404... array([[0, 1],
[2, 3],
[4, 5]])

```
In [405... arr.reshape(6,1)
```

Out[405... array([[0],
[1],
[2],
[3],
[4],
[5]])

```
In [406... arr.reshape(1,6)
```

Out[406... array([[0, 1, 2, 3, 4, 5]])

```
In [407... arr
```

Out[407... array([0, 1, 2, 3, 4, 5])

In [408... `arr.reshape(2,4)`

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[408], line 1  
----> 1 arr.reshape(2,4)  
  
ValueError: cannot reshape array of size 6 into shape (2,4)
```

In [409... `arr`

Out[409... array([0, 1, 2, 3, 4, 5])

In [410... `arr.reshape(2,3,order='C')`

Out[410... array([[0, 1, 2],
[3, 4, 5]])

In [411... `arr.reshape(2,3,order='F') # print element with fortran`

Out[411... array([[0, 2, 4],
[1, 3, 5]])

In [412... `arr.reshape(2,3,order='A') # A almost give you c type output`

Out[412... array([[0, 1, 2],
[3, 4, 5]])

In [413... `arr`

Out[413... array([0, 1, 2, 3, 4, 5])

In [414... `arr.reshape(2,3)`

Out[414... array([[0, 1, 2],
[3, 4, 5]])

In [415... `arr.reshape(1,4)`

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[415], line 1  
----> 1 arr.reshape(1,4)  
  
ValueError: cannot reshape array of size 6 into shape (1,4)
```

In [416... `arr.reshape(1,6)`

Out[416... array([[0, 1, 2, 3, 4, 5]])

```
In [417... arr.reshape(2,6)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[417], line 1  
----> 1 arr.reshape(2,6)  
  
ValueError: cannot reshape array of size 6 into shape (2,6)
```

```
In [418... arr.reshape(3,3)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[418], line 1  
----> 1 arr.reshape(3,3)  
  
ValueError: cannot reshape array of size 6 into shape (3,3)
```

```
In [419... arr
```

```
Out[419... array([0, 1, 2, 3, 4, 5])
```

```
In [420... arr.reshape(3,2)
```

```
Out[420... array([[0, 1],  
                [2, 3],  
                [4, 5]])
```

```
In [421... import matplotlib.pyplot as plt
```

```
In [422... from PIL import Image
```

```
In [423... butt_image=Image.open(r'C:\Users\HP\OneDrive\Pictures\butt.jpeg')
```

```
In [424... butt_image
```

```
Out[424...
```



```
In [425... type(butt_image)
```

```
Out[425... PIL.JpegImagePlugin.JpegImageFile
```

```
In [426... butt_arr=np.asarray(butt_image)  
butt_arr
```

```

Out[426... array([[[ 0,  4,  5],
                    [ 0,  4,  5],
                    [ 0,  4,  5],
                    ...,
                    [12, 28, 18],
                    [12, 28, 18],
                    [12, 28, 18]],

                  [[ 0,  4,  5],
                    [ 0,  4,  5],
                    [ 0,  4,  5],
                    ...,
                    [14, 30, 20],
                    [14, 30, 20],
                    [14, 30, 20]],

                  [[ 0,  3,  4],
                    [ 0,  3,  4],
                    [ 0,  3,  4],
                    ...,
                    [14, 30, 20],
                    [14, 30, 20],
                    [14, 30, 20]],

                  ...,

                  [[11, 22, 18],
                    [ 4, 15, 11],
                    [ 6, 17, 13],
                    ...,
                    [15, 25, 17],
                    [15, 25, 17],
                    [15, 25, 17]],

                  [[13, 24, 20],
                    [ 5, 16, 12],
                    [ 6, 17, 13],
                    ...,
                    [15, 25, 17],
                    [15, 25, 17],
                    [15, 25, 17]],

                  [[14, 25, 21],
                    [ 6, 17, 13],
                    [ 7, 18, 14],
                    ...,
                    [15, 25, 17],
                    [15, 25, 17],
                    [15, 25, 17]]], shape=(116, 175, 3), dtype=uint8)

```

```
In [427... plt.imshow(butt_image)
```

```
Out[427... <matplotlib.image.AxesImage at 0x2495db7d7f0>
```



```
In [428... butt_arr.shape
```

```
Out[428... (116, 175, 3)
```

```
In [429... butt_red = butt_arr.copy()
```

```
In [430... butt_red
```



```

Out[430... array([[ 0,  4,  5],
                  [ 0,  4,  5],
                  [ 0,  4,  5],
                  ...,
                  [12, 28, 18],
                  [12, 28, 18],
                  [12, 28, 18]],

                [[ 0,  4,  5],
                  [ 0,  4,  5],
                  [ 0,  4,  5],
                  ...,
                  [14, 30, 20],
                  [14, 30, 20],
                  [14, 30, 20]],

                [[ 0,  3,  4],
                  [ 0,  3,  4],
                  [ 0,  3,  4],
                  ...,
                  [14, 30, 20],
                  [14, 30, 20],
                  [14, 30, 20]],

                ...,

                [[11, 22, 18],
                  [ 4, 15, 11],
                  [ 6, 17, 13],
                  ...,
                  [15, 25, 17],
                  [15, 25, 17],
                  [15, 25, 17]],

                [[13, 24, 20],
                  [ 5, 16, 12],
                  [ 6, 17, 13],
                  ...,
                  [15, 25, 17],
                  [15, 25, 17],
                  [15, 25, 17]],

                [[14, 25, 21],
                  [ 6, 17, 13],
                  [ 7, 18, 14],
                  ...,
                  [15, 25, 17],
                  [15, 25, 17],
                  [15, 25, 17]]], shape=(116, 175, 3), dtype=uint8)

```

```

In [431... butt_arr == butt_red

```

```

Out[431]... array([[[ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True],
                    ...,
                    [ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True]],

                  [[ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True],
                    ...,
                    [ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True]],

                  [[ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True],
                    ...,
                    [ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True]],

                  ...,

                  [[ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True],
                    ...,
                    [ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True]],

                  [[ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True],
                    ...,
                    [ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True]],

                  [[ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True],
                    ...,
                    [ True,  True,  True],
                    [ True,  True,  True],
                    [ True,  True,  True]]], shape=(116, 175, 3))

```

```

In [432]... plt.imshow(butt_red)

```

```

Out[432]... <matplotlib.image.AxesImage at 0x2495dc34910>

```

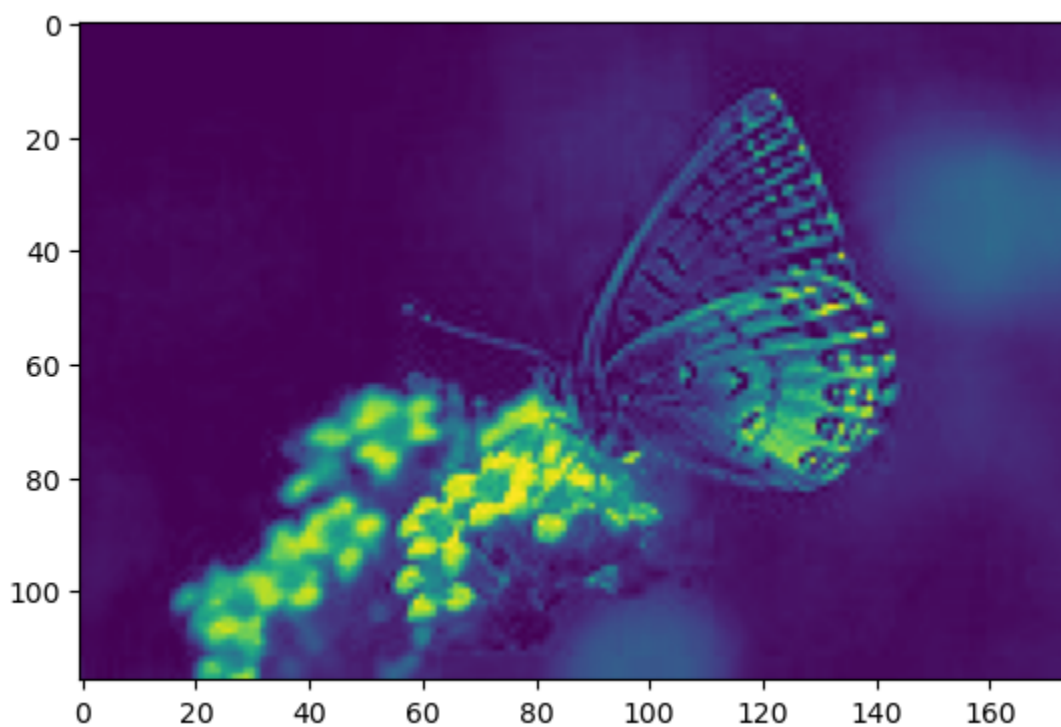


```
In [433... butt_red.shape
```

```
Out[433... (116, 175, 3)
```

```
In [434... plt.imshow(butt_red[:, :, 0])
```

```
Out[434... <matplotlib.image.AxesImage at 0x2495de0cb90>
```

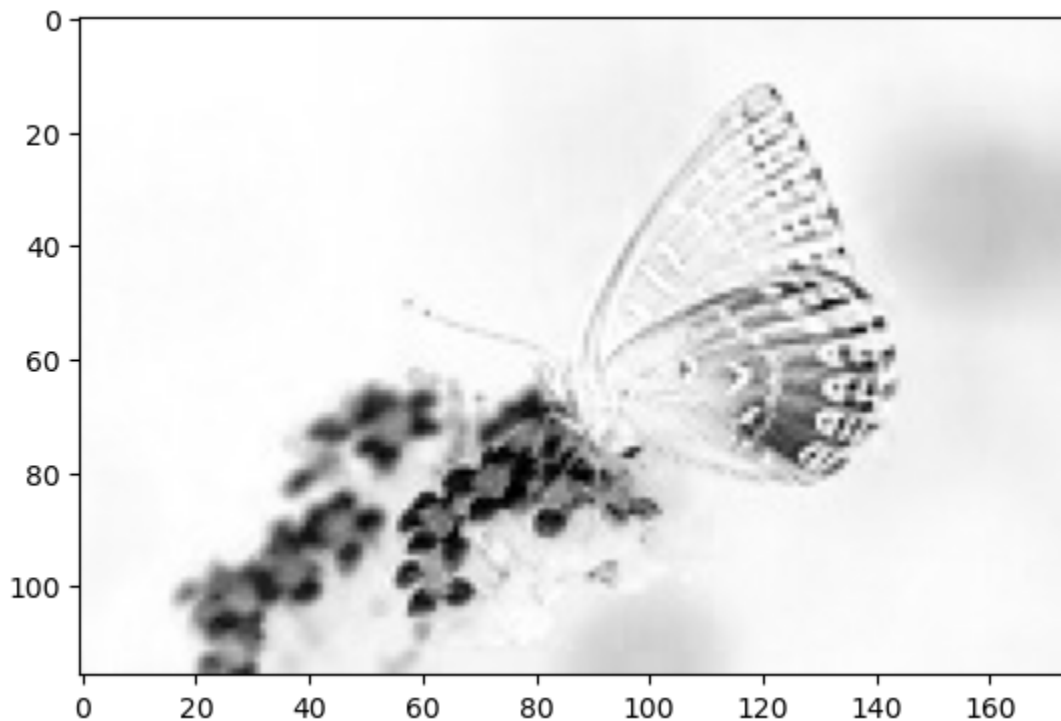


```
In [435... butt_red[:, :, 0]
```

```
Out[435... array([[ 0,  0,  0, ..., 12, 12, 12],  
        [ 0,  0,  0, ..., 14, 14, 14],  
        [ 0,  0,  0, ..., 14, 14, 14],  
        ...,  
        [11,  4,  6, ..., 15, 15, 15],  
        [13,  5,  6, ..., 15, 15, 15],  
        [14,  6,  7, ..., 15, 15, 15]], shape=(116, 175), dtype=uint8)
```

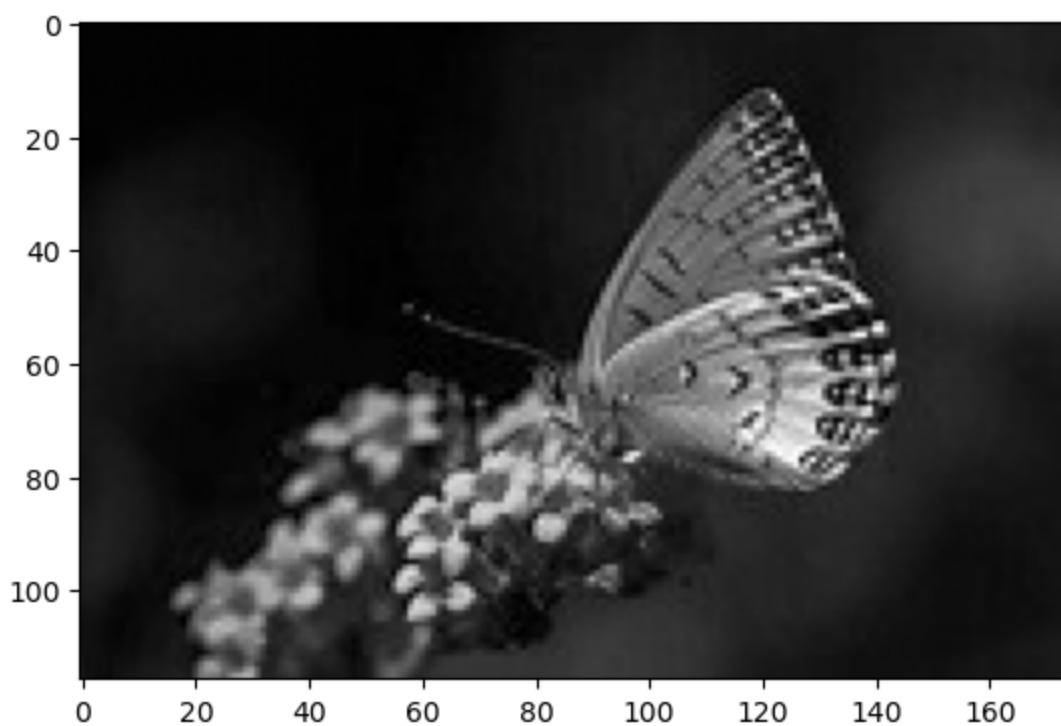
```
In [436... plt.imshow(butt_red[:, :, 0], cmap='Greys')
```

```
Out[436... <matplotlib.image.AxesImage at 0x2495de9c050>
```



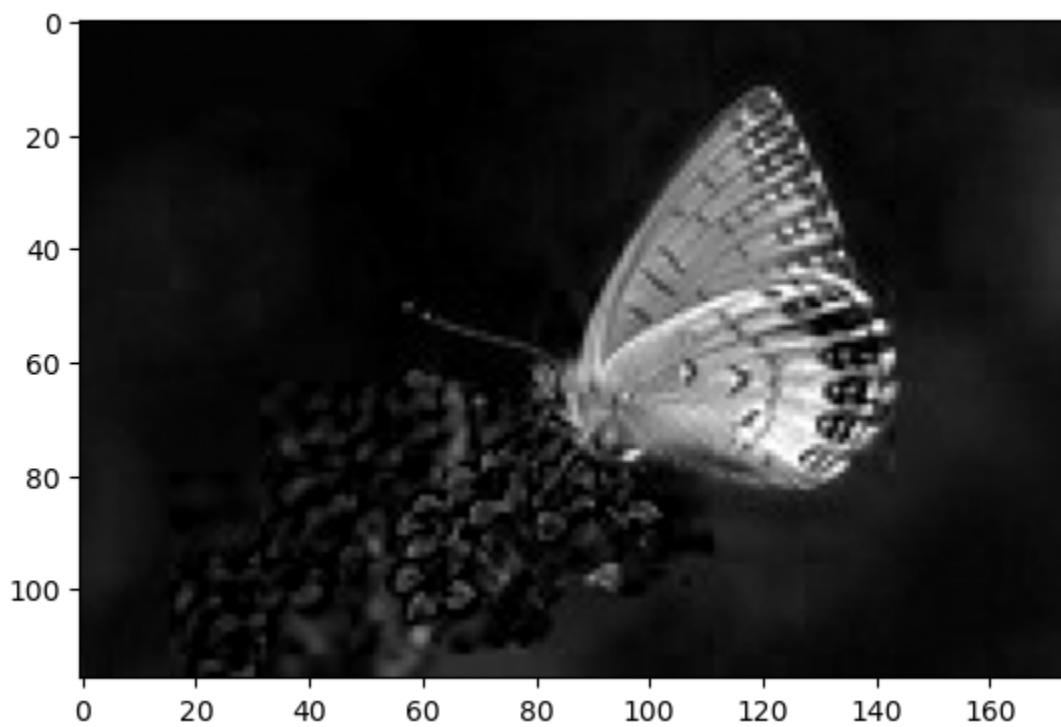
```
In [437... plt.imshow(butt_red[:, :, 1], cmap='grey')
```

```
Out[437... <matplotlib.image.AxesImage at 0x2495deeb610>
```



```
In [438... plt.imshow(butt_red[:, :, 2], cmap='grey')
```

```
Out[438... <matplotlib.image.AxesImage at 0x2495df82ad0>
```



```
In [439... butt_red[:, :, 0]
```

```
Out[439...] array([[ 0,  0,  0, ..., 12, 12, 12],
                  [ 0,  0,  0, ..., 14, 14, 14],
                  [ 0,  0,  0, ..., 14, 14, 14],
                  ...,
                  [11,  4,  6, ..., 15, 15, 15],
                  [13,  5,  6, ..., 15, 15, 15],
                  [14,  6,  7, ..., 15, 15, 15]], shape=(116, 175), dtype=uint8)
```

```
In [440...] butt_red[:, :, 1]
```

```
Out[440...] array([[ 4,  4,  4, ..., 28, 28, 28],
                  [ 4,  4,  4, ..., 30, 30, 30],
                  [ 3,  3,  3, ..., 30, 30, 30],
                  ...,
                  [22, 15, 17, ..., 25, 25, 25],
                  [24, 16, 17, ..., 25, 25, 25],
                  [25, 17, 18, ..., 25, 25, 25]], shape=(116, 175), dtype=uint8)
```

```
In [441...] butt_red[:, :, 2]
```

```
Out[441...] array([[ 5,  5,  5, ..., 18, 18, 18],
                  [ 5,  5,  5, ..., 20, 20, 20],
                  [ 4,  4,  4, ..., 20, 20, 20],
                  ...,
                  [18, 11, 13, ..., 17, 17, 17],
                  [20, 12, 13, ..., 17, 17, 17],
                  [21, 13, 14, ..., 17, 17, 17]], shape=(116, 175), dtype=uint8)
```

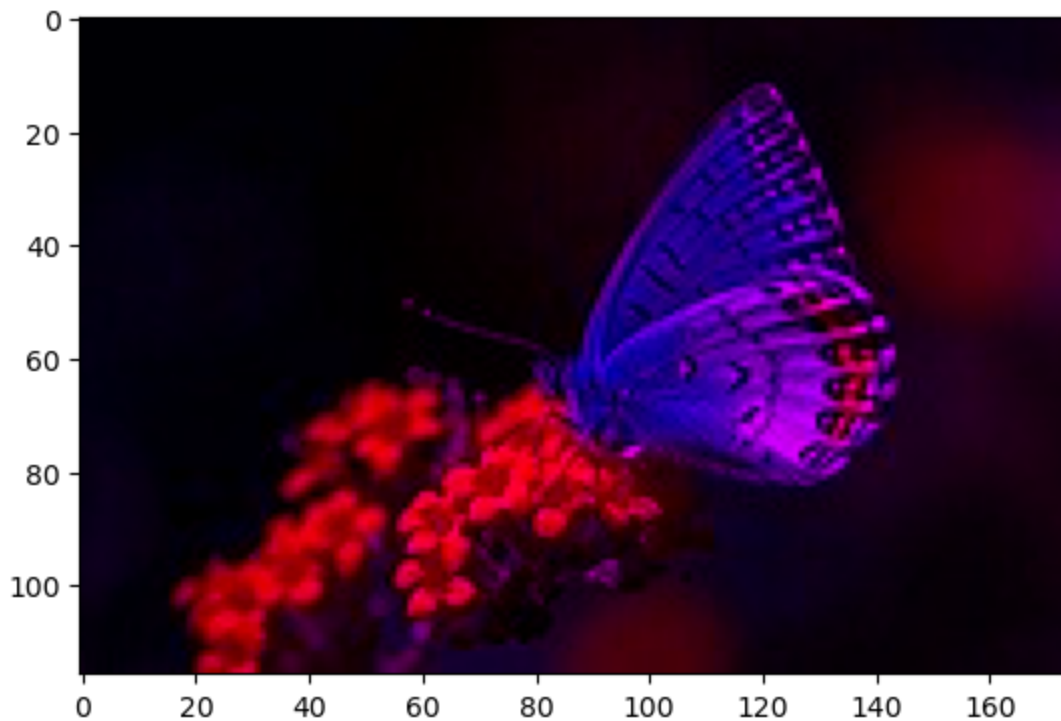
```
In [442...] butt_red[:, :, 1] = 0
```

```
In [443...] butt_red[:, :, 1]
```

```
Out[443...] array([[0, 0, 0, ..., 0, 0, 0],
                  [0, 0, 0, ..., 0, 0, 0],
                  [0, 0, 0, ..., 0, 0, 0],
                  ...,
                  [0, 0, 0, ..., 0, 0, 0],
                  [0, 0, 0, ..., 0, 0, 0],
                  [0, 0, 0, ..., 0, 0, 0]], shape=(116, 175), dtype=uint8)
```

```
In [444...] plt.imshow(butt_red)
```

```
Out[444...] <matplotlib.image.AxesImage at 0x2495e011e50>
```



```
In [445...] butt_red[:, :, 2]
```

```
Out[445...] array([[ 5,  5,  5, ..., 18, 18, 18],
 [ 5,  5,  5, ..., 20, 20, 20],
 [ 4,  4,  4, ..., 20, 20, 20],
 ...,
 [18, 11, 13, ..., 17, 17, 17],
 [20, 12, 13, ..., 17, 17, 17],
 [21, 13, 14, ..., 17, 17, 17]], shape=(116, 175), dtype=uint8)
```

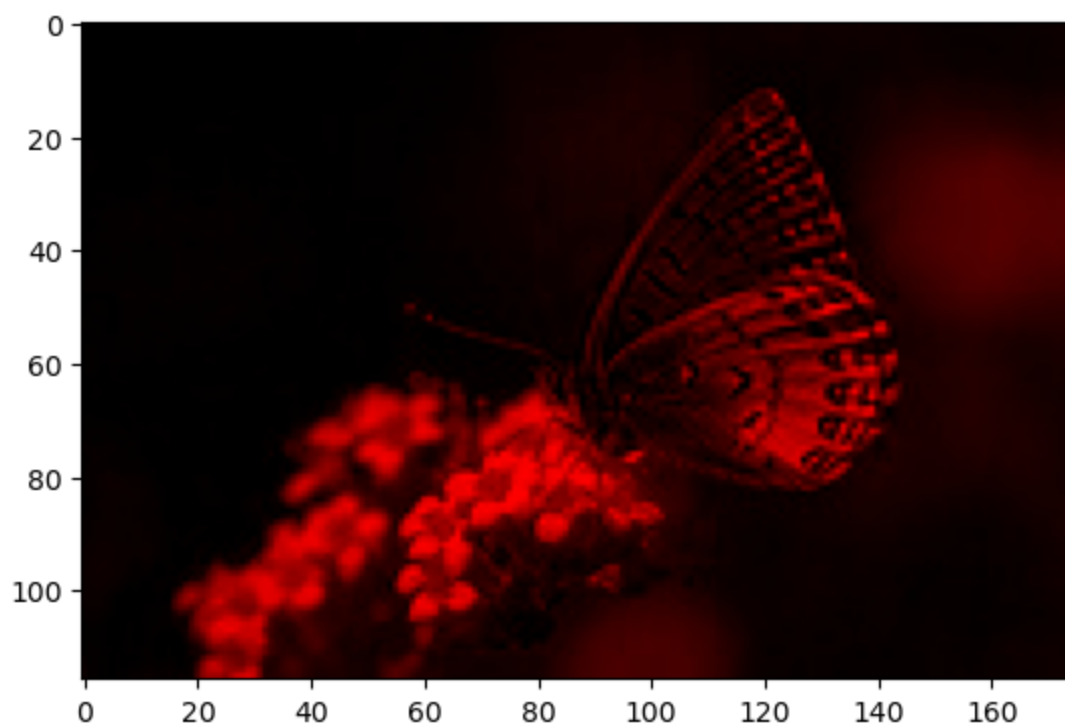
```
In [446...] butt_red[:, :, 2] = 0
```

```
In [447...] butt_red[:, :, 2]
```

```
Out[447...] array([[0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 ...,
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0],
 [0, 0, 0, ..., 0, 0, 0]], shape=(116, 175), dtype=uint8)
```

```
In [448...] plt.imshow(butt_red)
```

```
Out[448...] <matplotlib.image.AxesImage at 0x2495e08d310>
```



In [449... butt_arr


```

Out[449... array([[[ 0,  4,  5],
                    [ 0,  4,  5],
                    [ 0,  4,  5],
                    ...,
                    [12, 28, 18],
                    [12, 28, 18],
                    [12, 28, 18]],

                   [[ 0,  4,  5],
                    [ 0,  4,  5],
                    [ 0,  4,  5],
                    ...,
                    [14, 30, 20],
                    [14, 30, 20],
                    [14, 30, 20]],

                   [[ 0,  3,  4],
                    [ 0,  3,  4],
                    [ 0,  3,  4],
                    ...,
                    [14, 30, 20],
                    [14, 30, 20],
                    [14, 30, 20]],

                   ...,

                   [[11, 22, 18],
                    [ 4, 15, 11],
                    [ 6, 17, 13],
                    ...,
                    [15, 25, 17],
                    [15, 25, 17],
                    [15, 25, 17]],

                   [[13, 24, 20],
                    [ 5, 16, 12],
                    [ 6, 17, 13],
                    ...,
                    [15, 25, 17],
                    [15, 25, 17],
                    [15, 25, 17]],

                   [[14, 25, 21],
                    [ 6, 17, 13],
                    [ 7, 18, 14],
                    ...,
                    [15, 25, 17],
                    [15, 25, 17],
                    [15, 25, 17]]], shape=(116, 175, 3), dtype=uint8)

```

In [450... butt_red

```

Out[450... array([[[ 0,  0,  0],
                    [ 0,  0,  0],
                    [ 0,  0,  0],
                    ...,
                    [12,  0,  0],
                    [12,  0,  0],
                    [12,  0,  0]],

                  [[ 0,  0,  0],
                    [ 0,  0,  0],
                    [ 0,  0,  0],
                    ...,
                    [14,  0,  0],
                    [14,  0,  0],
                    [14,  0,  0]],

                  [[ 0,  0,  0],
                    [ 0,  0,  0],
                    [ 0,  0,  0],
                    ...,
                    [14,  0,  0],
                    [14,  0,  0],
                    [14,  0,  0]],

                  ...,

                  [[11,  0,  0],
                    [ 4,  0,  0],
                    [ 6,  0,  0],
                    ...,
                    [15,  0,  0],
                    [15,  0,  0],
                    [15,  0,  0]],

                  [[13,  0,  0],
                    [ 5,  0,  0],
                    [ 6,  0,  0],
                    ...,
                    [15,  0,  0],
                    [15,  0,  0],
                    [15,  0,  0]],

                  [[14,  0,  0],
                    [ 6,  0,  0],
                    [ 7,  0,  0],
                    ...,
                    [15,  0,  0],
                    [15,  0,  0],
                    [15,  0,  0]]], shape=(116, 175, 3), dtype=uint8)

```

In [451... butt_red

```

Out[451]: array([[[ 0,  0,  0],
                  [ 0,  0,  0],
                  [ 0,  0,  0],
                  ...,
                  [12,  0,  0],
                  [12,  0,  0],
                  [12,  0,  0]],

                 [[ 0,  0,  0],
                  [ 0,  0,  0],
                  [ 0,  0,  0],
                  ...,
                  [14,  0,  0],
                  [14,  0,  0],
                  [14,  0,  0]],

                 [[ 0,  0,  0],
                  [ 0,  0,  0],
                  [ 0,  0,  0],
                  ...,
                  [14,  0,  0],
                  [14,  0,  0],
                  [14,  0,  0]],

                 ...,

                 [[11,  0,  0],
                  [ 4,  0,  0],
                  [ 6,  0,  0],
                  ...,
                  [15,  0,  0],
                  [15,  0,  0],
                  [15,  0,  0]],

                 [[13,  0,  0],
                  [ 5,  0,  0],
                  [ 6,  0,  0],
                  ...,
                  [15,  0,  0],
                  [15,  0,  0],
                  [15,  0,  0]],

                 [[14,  0,  0],
                  [ 6,  0,  0],
                  [ 7,  0,  0],
                  ...,
                  [15,  0,  0],
                  [15,  0,  0],
                  [15,  0,  0]]], shape=(116, 175, 3), dtype=uint8)

```

In []:

In []:

In []:

In []:

In []: